



VISIBLE LIGHT OPTICAL TELESCOPE
Optical telescopes with the largest mirrors achieve the brightest, sharpest images and the greatest power (see p.82). They range from those of amateur astronomers, such as this example with an 8.5-in (21.5-cm) mirror, to those of large observatories, with mirrors up to 34 ft (10.4 m) wide. The Extremely Large Telescope (ELT) in Chile, with a 130-ft (39.3-m) wide mirror, will be the world's largest optical telescope once it is completed (expected 2024).



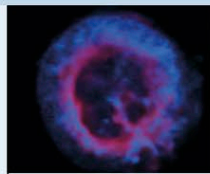
VISIBLE LIGHT GALAXY
The spiral galaxy M90, which lies 30 million light-years away, is shown here as it appears to human eyes through a large telescope. This galaxy is similar in size to the Milky Way. The image was taken at Kitt Peak National Observatory in Arizona.

EXPLORING SPACE
COMBINING THE VISIBLE AND INVISIBLE

Astronomers have developed telescopes that can gather information from EM radiation other than visible light, but they still face a problem of how to visualize the invisible. The most popular technique uses computers to create “false-color” images—pictures that show the object in particular wavelengths of radiation, sometimes color-coded, but often just using varying intensities of a single color. These images of a supernova remnant



1 Visible light image from Hubble Telescope and Very Large Telescope in Chile.

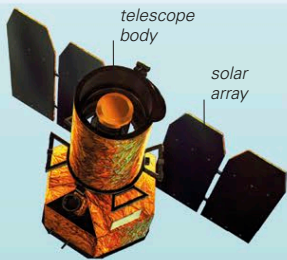


2 False-color X-ray image of E0102 taken by the Chandra X-ray Observatory.

known as E0102 show radiation in visible light and in X-rays, revealing the temperature of different regions. Combining the images shows the overall structure.



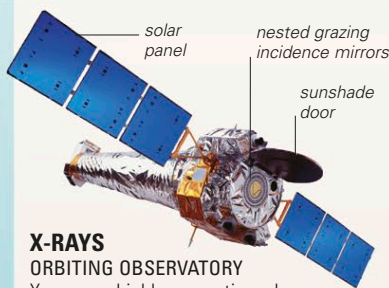
COMPOSITE IMAGE
Here, the visible light data has been combined with the false-color X-ray data.



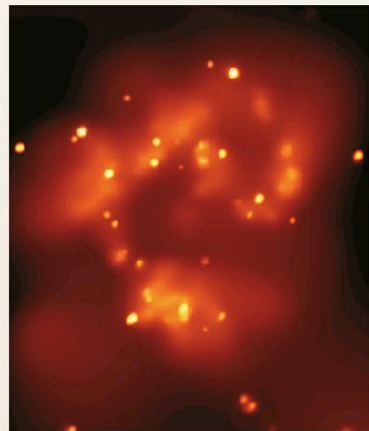
ULTRAVIOLET ORBITING OBSERVATORY
Between 2003 and 2012, NASA's Galaxy Evolution Explorer (GALEX) (above) surveyed sources of ultraviolet (UV) radiation. UV radiation originates from hot sources such as young stars, the Sun's corona, neutron stars, supernova remnants, and some galaxies.



ULTRAVIOLET SUPERNOVA REMNANT
Wispy tendrils of gas and hot dust glow brightly in this UV image of the Cygnus Loop supernova remnant, taken by GALEX. It is all that remains from a massive stellar explosion that occurred between 5,000 and 8,000 years ago.



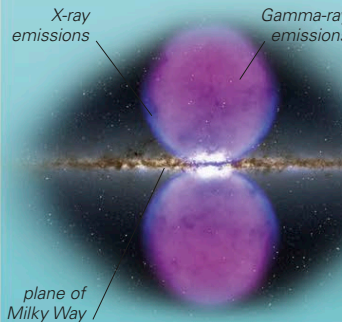
X-RAYS ORBITING OBSERVATORY
X-rays are highly energetic and so powerful that they pass through conventional mirrors. To focus X-rays, telescopes such as the Chandra X-ray Observatory (above) use a nest of curved “grazing incidence” mirrors of polished metal. X-rays glance off these mirrors, like ricocheting bullets, toward the focal point.



X-RAY GALAXY
The orange-pink regions in this Chandra Observatory image of two colliding galaxies (called the Antennae, see p.317) are X-ray-emitting “superbubbles” of hot gas. The point X-ray sources (bright spots) are black holes and neutron stars.



GAMMA RAYS ORBITING TELESCOPE
Gamma rays are the most energetic EM waves, emitted by the most violent cosmic events. The Fermi Gamma-Ray Space Telescope (above) was launched in 2008 to study gamma rays from phenomena such as supernovae, black holes, pulsars, and gamma-ray bursts.



GAMMA RAY GALAXY STRUCTURES
In 2010, using data from the Fermi Telescope, scientists discovered two gamma-ray-emitting lobes (shown here colored purple) extending for 25,000 light-years above and below the plane of our galaxy. These structures may be linked to a massive, energetic burst of star formation near the galaxy's center.

